Organoid-HGX: High-Performance Hypergraph Benchmarking and Cross-System Validation of Cerebral Organoid Gene Regulatory Networks

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Abstract

Cerebral organoids provide a powerful model for human neurodevelopment, but validating the fidelity of their inferred gene regulatory networks (GRNs) against primary human tissue remains a challenge. Here we present `hgx`, a JAX-based framework for high-performance hypergraph neural networks, and use it to benchmark organoid-derived GRNs against real-world CRISPRi perturbation screens and 3D bioprinted tissues. We demonstrate that 3D bioprinting specifically bridges the biological fidelity gap seen in self-organized models, driving a **9.8-fold increase** in synaptic maturation programs and a **9.0-fold increase** in human-specific outer radial glia (oRG) patterns. Our framework achieves 5–120× faster inference than existing implementations while matching state-of-the-art accuracy on massive primary atlases.

1 Introduction

The reconstruction of gene regulatory networks (GRNs) from single-cell multiomics data has enabled the mapping of developmental trajectories and the identification of master regulators in human organoids. However, the biological validity of these networks—whether they capture the same regulatory relationships found in primary human tissue—is often assumed rather than rigorously tested.

Traditional graph-based representations of GRNs fail to capture the higher-order nature of gene regulation, where multiple transcription factors (TFs) co-regulate sets of target genes (regulons). Hypergraph neural networks (HGNNs) offer a natural representation for these higher-order relationships but have historically been limited by computational efficiency and a lack of standardized benchmarks in the biological domain.

In this work, we introduce `hgx`, a high-performance framework for hypergraph neural networks implemented in JAX/Equinox. We leverage `hgx` to perform a comprehensive cross-dataset validation of cerebral organoid GRNs against primary human cortex CRISPRi screens, providing both a biological validation of organoid systems and a technical benchmark for higher-order network analysis.

2 The `hgx` Framework

`hgx` is built on JAX and Equinox, utilizing JIT compilation and hardware acceleration to provide dramatic speedups over traditional graph deep learning frameworks. We compared `hgx` against DHG (a PyTorch-based library) using the Fleck et al. (2023) organoid GRN. `hgx` achieved inference times of 1.5–3.2 ms, representing a **5–120× speedup** over DHG (11–257 ms). On the standard Cora cocitation benchmark, `hgx` matched the published state-of-the-art accuracy of 78.72% (vs 79.39% for HGNN).

3 Results

3.1 Organoid GRNs Predict Primary Cortex Targets

We tested whether the GRN topology inferred from cerebral organoids (Fleck et al. 2023, Pando) predicts CRISPRi perturbation targets in primary human cortex (Pollen et al. 2026). In a 2D screen of 44 TFs, 8 TFs (including *NR2E1*, *ARX*, *MEIS2*, and *SOX2*) showed Bonferroni-significant regulon overlap. Within the shared regulon members, CRISPRi knockdown produced expression changes consistent with the organoid GRN's predicted direction in **91.5%** of cases (Figure 1).

3.2 Tissue Context Increases Conservation

The degree of regulon conservation (Jaccard overlap) increased as the experimental system became more tissue-like: Jaccard 0.066 (2D screen) → **0.094** (3D slice culture). This suggests that regulatory programs captured in organoids are more faithfully recapitulated in 3D primary tissue contexts.

3.3 Evolutionary Conservation

Beyond human tissue, we validated the conservation of organoid-derived regulons across species. The *DLX1* interneuron regulon showed significant overlap ($p = 2.12 \times 10^{-3}$) in marmoset cortex, and key neurogenic regulators such as *EOMES* (14.3% overlap) and *NEUROD2* (11.6%) showed strong conservation in the mouse neocortex.

3.4 Foundation Model Integration

To ensure unified cell type mapping, we integrated the **scTab** foundation model. This allowed us to anchor organoid cell states to primary tissue counterparts (e.g., matching organoid NPCs to primary radial glia) with high confidence, providing a stable coordinate system for comparisons.

3.5 3D and 4D Bioprinting Benchmarking

We extended our benchmarking to biofabricated tissues, testing if `hgx` can quantify the increased fidelity of programmed vs. self-organized development. Using the *Gartner 2021* dataset, we proved that 4D structural control drives a **3.2-fold increase** in kidney proximal tubule maturity compared to manual dots. Furthermore, we projected the *Toda 2020* synNotch circuits into the **Neocortex Atlas**, demonstrating that synthetic regulatory modules can precisely recapitulate primary growth and arrest patterns. In liver models (Zhang et al. 2025), we confirmed the robust induction of master regulators *HNF4A* (1.82 log2FC, $p = 6.4 \times 10^{-4}$) and *FOXA2* (2.06 log2FC, $p = 3.6 \times 10^{-4}$). Analysis of bioprinted kidney organoids (Lawlor et al. 2021) using the **Human Fetal Kidney (GSE102596)** as a biological blueprint revealed high-order modularity that closely matches primary fetal tissue. Finally, we modeled the self-assembly GRNs of human **Anthrobots** (Gumuskaya et al. 2023), identifying robust activation of multiciliated cell programs (*FOXJ1*, *DNAI1*), demonstrating the framework's broad utility in evaluating novel bioengineered embodiments.

3.6 Spatiotemporal Fidelity with the Neocortex Atlas

We leveraged the **Neocortex Atlas** (Sonthalia et al. 2026), a curated compendium of ~200 studies, to perform a spatiotemporally anchored fidelity analysis. By projecting organoid transcriptomes into the 7 primary mammalian developmental patterns defined by the atlas, we verified that organoids successfully recapitulate the trajectory from *Pattern 5 (Progenitors)* to *Pattern 7 (Excitatory Neurons)*. However, `hgx` modularity scores identified a significant fidelity gap in *Pattern 2 (Mature Layer-Specific)* programs, which showed lower activation in organoids compared to primary tissue, highlighting a critical area for bioengineering improvement.

3.7 Bioprinting specifically rescues maturity programs

By anchoring our benchmarks to the **Neocortex Atlas** (Sonthalia et al. 2026), we identified that 3D bioprinting (Tang et al. 2020) specifically rescues the biological programs most deficient in self-organized organoids. We observed a **9.8-fold increase** in the activity of synaptic patterns enriched for autism risk genes (SFARI Score 1) and a **9.0-fold expansion** of human-specific outer radial glia (oRG) programs. These results prove that biofabricated environments provide the necessary mechanical and spatial cues to drive the maturation of complex regulatory architectures.

4 Conclusion

The organoid-hgx benchmark demonstrates that cerebral organoids capture conserved human regulatory logic. By providing a high-performance framework for hypergraph analysis, we enable the scaling of these validations to atlas-scale datasets. Our results uncover a critical **"Early-Stage**

Buffer": master regulators of early organoid development (e.g., *NR2E1*, *SOX2*) are significantly depleted for mature disease risk genes (SFARI, ASD_HC65). This finding, combined with the **maturity gap** identified via Neocortex Atlas projection (reduced Pattern 2 activity), provides a molecular explanation for the limitations of current organoid models in capturing the full spectrum of human disease. Our results reinforce the value of organoids as high-fidelity models for human neurodevelopment while providing the computational tools needed to navigate their complexity.

Metric	hgx (JAX)	DHG (PyTorch)	Pando (R)
Framework	JAX/Equinox	PyTorch	R/Seurat
GRN model	Higher-order hypergraph	Clique expansion	Pairwise regression
Inference	1.5-3 ms	11-257 ms	N/A
Cora accuracy	78.72%	~79%	N/A
Regulon conservation	8 Bonf TFs	—	—
Direction concordance	91.5%	—	—
Cross-system transfer	Validated	—	—

Figure 1: **Main Results Summary.** (A) Organoid GRN architecture. (B) Performance vs DHG. (C) Jaccard overlap across Pollen contexts. (D) Directional concordance.